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Development and testing of an efficient data acquisition platform for machine learning of optical emission spectroscopy of plasmas in aqueous solution

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Abstract

This study presents the development of an efficient data acquisition platform and discusses machine learning of optical emission spectroscopy (OES) of plasma in aqueous solution to demonstrate the multivariate analysis of spectra. A specially designed platform for efficient acquisition of spectra emanated from plasmas in solutions is developed, and several machine learning algorithms are tested for plasma analysis. This platform enables acquiring up to 10k spectra solutions with various pH values under constant solution conductivity, or vice versa, within 15 min. This rapid acquisition scheme provides a sufficient dataset for testing machine learning algorithms. We test the OES of plasmas ignited in solutions with designated conductivities with pH of 2.2–5.2. A total 40k spectra are collected and tested with principal component analysis (PCA) and an artificial neural network (ANN) to predict the conductivity of the solution. In PCA, the results show that most data points are overlapped in the score plot constructed using principal components 1 and 2, implying that PCA cannot discriminate the conductivity based on the spectra. In an ANN, several network structures are constructed and tested. The results show that the deep ANN significantly improves the accuracy of conductivity prediction in terms of mean squared error by three orders of magnitude compared with the method of using single emission line. Regularization techniques including dropout and early stopping show promise for mitigating overfitting in a deep ANN. Such improvements suggest that a deep ANN considers the nonlinear behaviors of plasma and can handle datasets with high complexity. In addition, the application of a deep ANN with a large number of parameters makes using this experimental platform for efficient spectra acquisition highly desirable.

Supplementary material for this article is available online

Keywords: machine learning, optical emission spectroscopy, plasma-liquid interaction, solution plasma, principal component analysis, artificial neural network

1. Introduction

Optical emission spectroscopy (OES) of plasmas is the most widely used tool for plasma diagnostics [1]. OES provides useful information about plasma characteristics and the existence of certain peaks can be used to identify specific excited species generated in plasmas. It allows for qualitative and quantitative analyses of both the plasma and its surrounding environment. It also enables real-time and non-invasive monitoring of processes for various applications such as materials synthesis [2, 3], environmental remediation [4–6], and analytical chemistry [7]. Conventionally, one or few wavelengths are extracted from OES to analyze the plasma characteristics. However, this approach is not sufficient to fully characterize the plasma for the following reasons. (1) The plasma is highly nonlinear in nature and its OES is a complicated function of operating conditions and plasma characteristics. (2) The emission intensities at different wavelengths are highly coupled with complicated plasma dynamics [8]. (3) OES variations exist even under identical operating conditions [9]. Therefore, one or few emission peaks carry very limited information. Owing to the above-mentioned problem, full spectrum analysis is strongly desired.

Among the various plasma sources, plasma ignited in a solution has attracted considerable attention because of its numerous applications [10–12]. The development and characterization of such plasma systems with various setups have been demonstrated [13–17]. The analysis of the plasmas of such systems has been challenging owing to the existence of complicated plasma–electrode–liquid interactions. Because plasmas are submerged in liquids, their behavior is temporally and spatially non-uniform, making OES analysis challenging [18].

In recent years, studies have demonstrated the effectiveness of machine learning for spectroscopy [19]. It can perform multivariate analysis of spectra and extract valuable predictive information from large datasets through data mining [20]. Principal component analysis (PCA), an unsupervised machine learning technique, of OES has been widely used for endpoint detection of plasma etching processes in the semiconductor industry [21–23]. PCA determines the variance of the dataset and can be used for dimension reduction, thereby providing more accurate information about the endpoint. Through full spectrum analysis, PCA can more precisely identify the etching endpoint for wafers with a smaller etching area. However, PCA has several limitations [24]. First, it is a linear dimension reduction method and may therefore miss the features of datasets with nonlinear and non-orthogonal patterns. Second, it may provide biased information owing to the fluctuating behavior of plasma rather than the patterns of interest. To address these issues, supervised machine learning methods are required for spectroscopy analyses. The use of algorithms such as artificial neural network (ANN) [25, 26], support vector machine [27–29], and k-nearest neighbors [30], has been suggested for this purpose. Utilizing ANN for endpoint detection has been reported in several studies [31–33].

When using machine learning for spectroscopy analyses, the veracity of the training data is typically an important factor [34, 35]. A sufficient number of representative spectra (typically, $\geqslant$10 000) is required to ensure the trustworthiness of predictions from machine. The plasma behavior and optical emission are sensitive to experimental conditions such as the instruments, system schematic, gas and solution compositions, and operating parameters. It is difficult to use a general database for training on an individual experiment. Therefore, a customized spectrum generation platform enabling rapid acquisition of plasma OES to construct a database for machine learning is essential.

In this study, we developed a platform that enables efficient acquisition of plasma OES using plasma ignited in a solution. This platform integrates a plasma generation unit and a spectrometer with a solution feeding scheme. This platform allows for continuous acquisition of plasma OES while the solution property, namely conductivity or pH, is continuously changing. We test several machine learning schemes, namely, PCA and ANN, for plasma OES analysis. We quantify the solution conductivity with various pH values using plasma OES because the conductivity of the solution is one of the critical properties in the applications of material synthesis, biomedical treatment, and environmental remediation using plasmas in a solution. This experimental platform can efficiently acquire plasma OES for plasmas ignited in a solution with a designated solution conductivity and various pH values. A total of 40k spectra are collected within tens of minutes. We performed tests using a single emission line, PCA, and shallow and deep ANN to predict the solution conductivity. The accuracy of conductivity prediction for each scheme was discussed. We also discussed the implications and perspectives of using machine learning for plasma spectroscopy as a route for characterizing plasmas.

2. Material and methods

2.1. Plasma OES generation and acquisition platform

Figure 1(a) shows a schematic of the platform for the efficient acquisition of plasma OES. This platform consists of a plasma generation unit, a pumping unit, and a spectrometer. The plasma generation unit generates plasmas in a solution. It consists of a glass cell with a discharge and a grounding electrode. The discharge electrode is a 0.5 mm diameter platinum wire and is covered by a glass tube to restrict the area in contact with the solution. The grounding electrode is a bare platinum wire with the same diameter. To ignite the plasmas, the discharge electrode is driven by a pulsed power source consisting of a DC power supply (GWinsek, GPR-100H15) and a switching device. A transistor controlled by a microcontroller (Arduino, Arduino Uno) is used as the switching device. We apply repetitive DC pulsed waveforms with a voltage of 800 V and a pulse width of 4 ms to ignite the plasma. The pulse frequency is 50 Hz. Typical voltage and current waveforms are shown in the supporting information (see figure S1 available online at stacks.iop.org/PSST/28/105013/mmedia). To acquire plasma OES, an optical emission spectrometer (Ocean Optics, USB 2000) is used. Integration times of 25–800 ms are used to acquire spectra with various signal-to-noise ratios. The pumping unit consists of a peristaltic pump; it is used to continuously inject solutions with a flow rate of 50 ml min^{-1} into the solution in which plasma is generated. With proper selection of the solution in the plasma generation unit and the solution that is pumped into this unit, plasmas can be generated in the solutions under various conditions. Plasma OES can be acquired. With the integration of the plasma generation unit, the spectrometer, and the solution feeding scheme, plasma OES can continuously be acquired while the solution property, namely conductivity or pH, is continuously changing. Up to 10k spectra can be acquired in plasmas ignited in solutions with various pH values under designated conductivity, or vice versa, within 15 min. A total of 40k spectra are collected

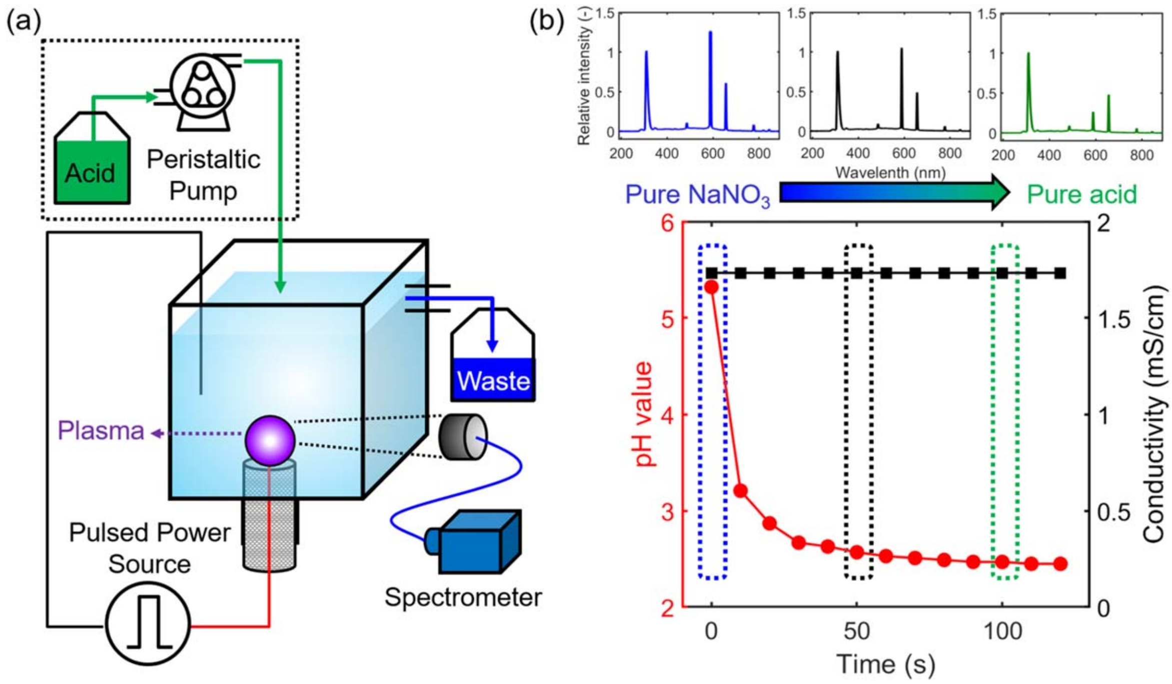

Figure 1. (a) Schematic of plasma OES generation platform. (b) pH and conductivity profiles in solutions during the injection of pure nitric acid into pure $NaNO_3$. The insets show the spectra acquired at 0, 50, and 100 s, respectively.

within tens of minutes using this platform. The following section discusses the experimental conditions in detail.

2.2. Structure of dataset

We use this platform to generate an extensive number of spectra for machine learning. These spectra are emanated from plasmas generated in solutions with conductivity of 500–2300 μS cm^{-1}. Two chemicals, $NaNO_3$ (Honeywell, 99.5%) and nitric acid (J.T. Baker, 69.5%), are used to adjust the conductivity and pH of the solution because they are typical contaminants in wastewater. Solutions with five different conductivities are prepared. The test solution contains mixtures of $NaNO_3$ and nitric acid with different fractions. We note that the plasma characteristics and its optical emission vary with the $NaNO_3$ and nitric acid fractions in the solutions even under identical conductivity. A dataset with 40k spectra is constructed. These spectra consist of 8k spectra emanated in solutions with each conductivity. Of these, 3k, 3k, and 2k spectra are acquired in pure $NaNO_3$ solutions, pure nitric acid solutions, and a mixture of $NaNO_3$ and nitric acid, respectively. For spectra emanated in pure $NaNO_3$ and pure nitric acid solutions, we continuously drive the plasma and collect data using the spectrometer. The solution is continuously pumped into and drained out of the plasma generation unit to maintain the solution temperature. The 3k spectra are collected within 10 min. For spectra generated in the mixture of $NaNO_3$ and nitric acid, pure nitric acid solutions are continuously pumped into the plasma generation unit initially containing pure $NaNO_3$ solution. This approach yields a mixture of solutions with various pH values under a constant conductivity when pure $NaNO_3$ and pure nitric acid solutions with identical conductivity are used. Figure 1(b) shows the temporal profiles of conductivity and pH during the mixing process. We continuously ignite plasma and collect spectra during the pumping process, which enables us to acquire spectra with various solution conditions efficiently. The insets in figure 2(b) show spectra collected from a solution with a conductivity of 1733 μS cm^{-1} and pHs of 5.15, 2.44, and 2.24, corresponding to 0, 50, and 100 s. The 2k spectra are collected during the pumping process within 5 min. A total of 40k spectra are acquired in solutions with various conductivities and pH values within tens of minutes. Table 1 lists the number of spectra and the corresponding solution conditions.

The spectra are preprocessed before performing machine learning. The Na emission line at 589.6 nm, which is the most prominent Na emission in the spectra, is filtered in each spectrum to eliminate the bias caused by the Na concentration and the saturation of Na intensity. In addition, we normalize the intensities to the OH emission line at 306.3 nm in each spectrum to avoid the bias resulting from different light intensities collected for different sets of spectra. The OH emission line is chosen because it is one of the most prominent emission lines in the plasma-liquid system, as reported previously [8]. After preprocessing, each spectrum consists of 1909 dimensions, that is, emission intensities at 1909 different wavelengths.

2.3. PCA

PCA is an unsupervised multivariate machine learning method that is increasingly used for spectroscopy analyses [36–38]. PCA considers the full spectrum as a multidimensional vector where each wavelength in the spectrum represents one dimension. PCA identifies the linear

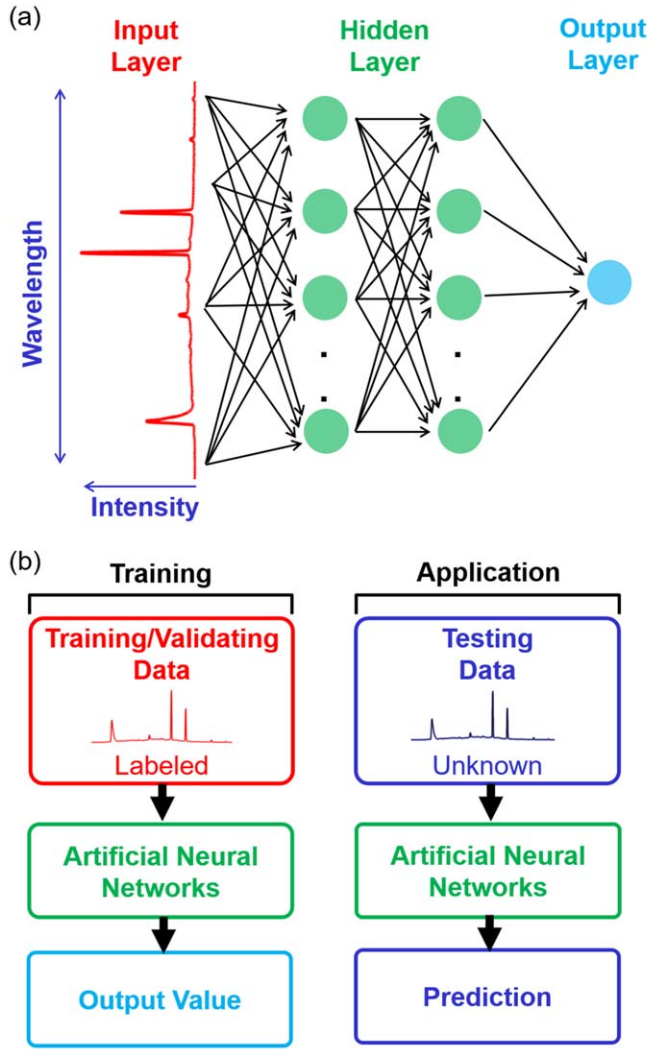

Figure 2. (a) Illustration of fully connected ANNs for dealing with spectra and (b) conceptual diagram of the data flow for training and application.

Table 1. Number of spectra in dataset with corresponding solution conditions.

Conductivity (μS cm^{-1})	Pure $NaNO_3$ solution	Mixture of $NaNO_3$ and HNO_3	Pure HNO_3 solution
492	3k	2k	3k
909	3k	2k	3k
1345	3k	2k	3k
1733	3k	2k	3k
2325	3k	2k	3k

combination of each dimension of the spectrum that shows the variance among multiple spectra. The linear combinations that show the highest, second-highest, and third-highest variance are the first, second, and third principal components (PCs), respectively. This technique considers the full spectrum to describe the features in the dataset using a small number of PCs. By performing PCA, the complexity of the dataset, that is, the spectrum, can be simplified. When finding PCs, each PC is orthogonal to prevent the overlap of features. The loading plot of each PC allows for understanding the importance of each dimension that contributes to the PC. It enables the characterization of the major features of each spectrum. In this study, our goal is to use PCA to capture the spectral variance for quantifying the conductivity of test solutions. The 40k sets of spectra are applied to the PCA with each spectrum containing 1909 dimensions, that is wavelengths. The *pca* function in Matlab is used to perform PCA.

2.4. ANN

ANN is a supervised machine learning scheme that has shown great success in various fields in recent years [39–41]. ANNs can perform complex calculations using inputs with little or no feature engineering. In this study, we explore fully connected ANN. Our goal is to train a framework that can precisely quantify the conductivity of a solution with various pH values using the OES. Figures 2(a) and (b) respectively show an illustration and conceptual diagram. The ANN consists of three different types of layers: input layer, hidden layer, and output layer [42]. Each layer is constructed with several nodes, namely, neurons. The input layer is constructed using experimentally obtained spectra. Each spectrum contains 1909 wavelengths, corresponding to 1909 neurons. The hidden layer takes signals from the input layer and performs linear or nonlinear transformations. A rectified linear unit (ReLU) nonlinear activation function is used in each neuron in the hidden layer [43]. Multiple hidden layers with different numbers of neurons can be stacked to build up ANNs with various architectures. The output layer performs linear transformations on the signals from the previous layer and provides predictions of the solution conductivity. The ANN parameters are learned and optimized using a back-propagation algorithm with the Adam update scheme [44, 45]. The mean square error (MSE) loss function is used to evaluate the predictions. The MSE equation is as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{\mathrm{i}}(\mathrm{y_i} - \hat{\mathrm{y}}_{\mathrm{i}})^2$$

where y_i is the predicted value and $\hat{y}_i$, the actual measured value. In addition, root mean square error (RMSE), the square root of MSE, is also used to evaluate the prediction. For testing the model, three ANNs with different structures (i.e., different numbers of hidden layers and neurons) are constructed and compared: (1) ANN with zero hidden layers, (2) shallow ANN with one hidden layer containing 50 neurons, and (3) deep ANN with three hidden layers containing 2000, 50, and 10 neurons in the 1st, 2nd, and 3rd hidden layers, respectively.

When using the ANN method, one typical concerning issue is overfitting. Overfitting is a situation in which the loss evaluated using training data is much lower than that evaluated using test data [46]. Overfitting commonly appears when using a large number of parameters to train the ANN. To alleviate overfitting, we randomly divide the dataset into a training set, a validation set, and a test set. In addition, strategies including k-fold cross validation (k = 5), dropout, and early stopping are implemented. The training and validation sets contain 95% of the spectra in the dataset and are used for training of the ANN. The test set containing the remaining 5% of the dataset is used to provide an unbiased evaluation of the final trained framework. We note that the test set is not used during the training process. It is only used to evaluate the trained framework after the training process is finished. In k-fold cross validation, we split the training and validation sets into five groups. Of these, four are used as the training set and one is used as the validation set in each training step. The training set is used for optimizing the parameters in the ANN and the validation set, for providing an unbiased evaluation of the results obtained using the training set. We note that the validation set is independent of the training set while updating the parameters. By comparing the results between the training and the validation sets, whether or not overfitting occurs can be justified and the hyperparameters of the ANNs can be tuned. Moreover, we train each ANN five times with different compositions of training and validation sets to ensure the robustness of the framework. Dropout and early stopping are implemented during the training of the deep ANN to mitigate overfitting. Dropout is a typical regularization technique for reducing overfitting in the ANN [46]. Dropout randomly masks several neurons in each iteration and efficiently performs averaging to the parameters. A 10% dropout rate is applied in the first hidden layer of the deep ANN. We note that the dropout rate and the layer in which to perform dropout are optimized. Early stopping is another technique that is widely used in ANNs [47]. This techniques is aimed at achieving a lower MSE in the validation set rather than in the training set. Early stopping with a tolerance of 80 epochs is employed. During iterations, the machine automatically terminates the training stage when a lower MSE in the validation set is not acquired for 80 epochs. Although the accuracy of the training set may be sacrificed, it minimizes the generalization error and ensures accuracy for future prediction.

To set up and train the ANN, TensorFlow and Keras are used. In addition, CUDA and GTX 1066Ti are used to accelerate the training process. The time required for the training process of each ANN is within 2 h.

3. Results and discussion

3.1. Representative OES and quantification of solution conductivity

Figures 3(a)–(e) show typical emission spectra of plasma ignited in pure $NaNO_3$ solutions with a conductivity of 492–2325 μS cm^{-1}. Each spectrum represents the average of 3k spectra. The intensities are normalized to the emission intensities of OH at 306.3 nm. The emission line of Na (589.6 nm) is removed because solutions with different $NaNO_3$ concentrations are used. In addition, its intensity is much higher than that of other lines and is saturated in several spectra. The emission lines of H_α, H_β, O, and OH are

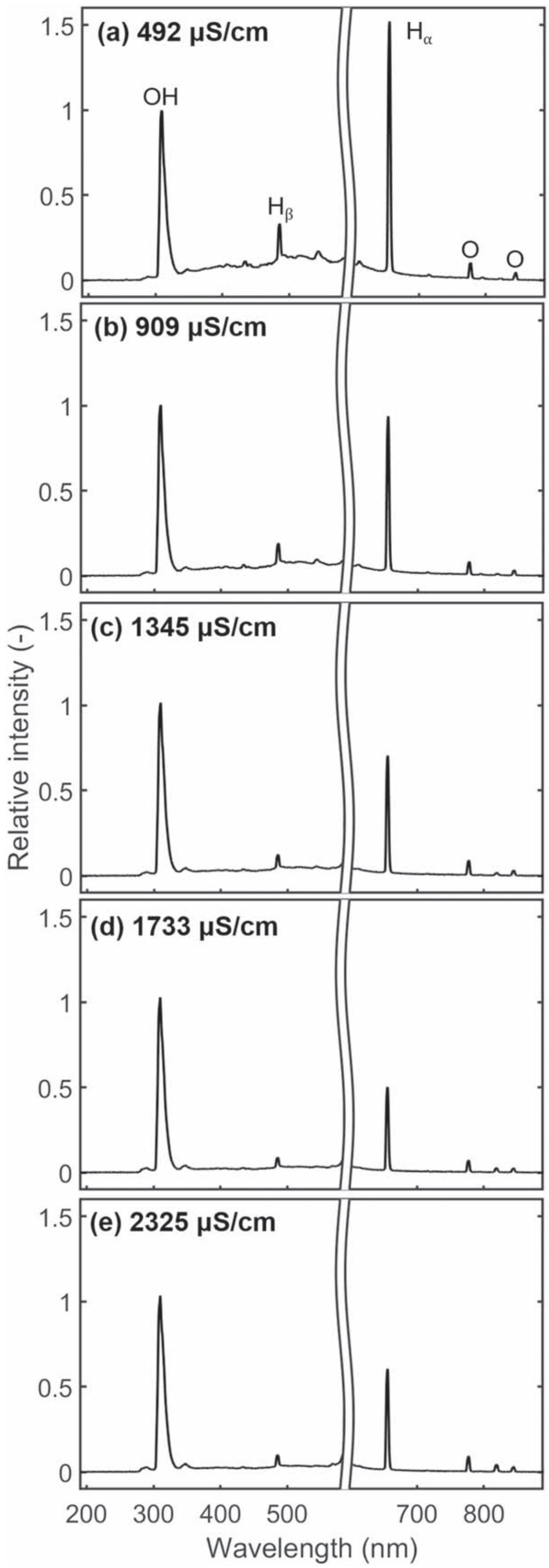

Figure 3. Representative OES generated in $NaNO_3$ solutions with conductivities of (a) 492, (b) 909, (c) 1345, (d) 1733, and (e) 2325 μS cm^{-1}. The characteristic peak of Na at 589.6 nm is removed in each spectrum.

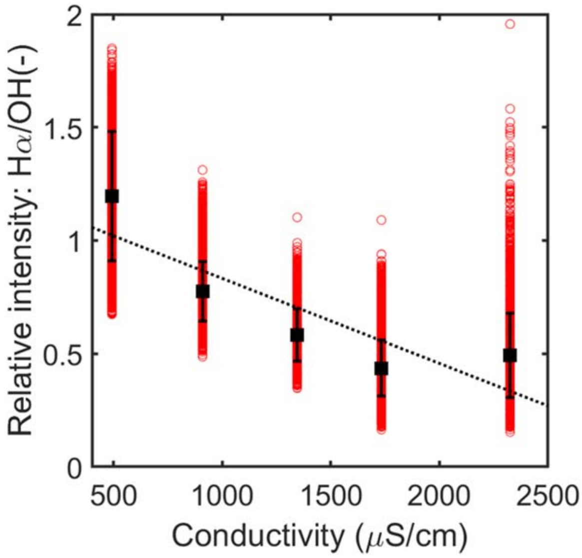

Figure 4. Relationship of the relative intensity of H_α/OH and the conductivity of the solution. The red points indicate the values of each spectrum. The black marks indicate the average values and their SD. The dotted line indicates the fitted line with first-order linear regression.

observed. We also observed that the emission intensity ratio of H_α/OH decreases with increasing conductivity in the solutions. This observation is consistent with those reported in literature concerning observations of H_α, H_β, O, and OH emission lines in plasmas ignited in aqueous solutions [8, 48–51]. With the application of a high voltage, H_2, O_2, and water vapor are generated through electrolytic reactions and Joule heating. Plasma formation leads to multiple complex reactions such as electron impact ionization, dissociation, and excitation. These reactions are responsible for the occurrence of H_α, H_β, O, and OH emission lines.

We test the capability of conductivity quantification by observing the H_α emission line. Figure 4 shows the results for quantifying conductivity in the range of 500–2300 μS cm^{-1}. The 8k spectra acquired from the plasma were ignited in a pure $NaNO_3$ solution, pure nitric acid solution, and mixtures of $NaNO_3$ and nitric acid with various pH values for each conductivity. Each red point presents the H_α/OH value of each spectrum. The averaged value and standard deviation (SD) are indicated by black marks. The dotted line indicates the fitted line with first-order linear regression. The results show that the averaged H_α/OH value decreases when conductivity increases from 500–1750 μS cm^{-1} and slightly increases at 2300 μS cm^{-1}. We also observe a large deviation within each conductivity with an MSE of 2.31×10^5. This observation clearly shows that the first-order regression line cannot accurately represent the conductivity of solutions. We further observe spectra that show the best, average, and the worst accuracy according to the fitted line. Figure 5 shows the three spectra in solutions with each conductivity tested from 500–2300 μS cm^{-1}. We note that the intensities for each spectrum are normalized by the intensity of the OH emission intensity at 306.3 nm. The spectra with H_α/OH values equal to the best fitted value, averaged value, and value with the largest deviation are shown. Spectra with significant difference are observed not only in the H_α/OH values but also in other emission line intensities. It clearly shows the large variation of spectra with each conductivity; this is attributed to a rather complicated interaction among the gas, solution, and electrode. Therefore, the normalized intensity of the H_α emission line cannot be used to accurately predict solution conductivity.

This poor prediction accuracy results from the characteristics of plasma in different solutions and the non-uniform nature of the plasma. Plasmas ignited in solutions are sensitive to the solution compositions. Even for solutions with similar conductivity, the presence of various types and compositions of metallic salts and acids may lead to different current waveforms, temperature, and bubble dynamics behaviors and therefore influence the optical emission. In addition, plasma generation is highly non-uniform spatially and temporally. These synergetic effects make conductivity quantification using one or few emission lines challenging.

3.2. Feature selection using PCA

Next, we apply PCA to the dataset containing 40k spectra. PCA is one of the most commonly used machine learning algorithms for plasma OES in various applications such as endpoint detection in plasma etching processes in the semiconductor industry. PCA identifies the PCs, that is, vectors along which the maximum variation among datasets occurs. We examine whether the features (i.e., PCs) identified by PCA can interpret the difference among solutions with various conductivities. Figure 6(a) shows the results through a two-dimensional score plot constructed by PC1 and PC2. We indicate data points with different colors for different conductivities after performing PCA. PC1 and PC2 capture 91.6% and 4.2% of the variance, respectively. The percentage of variance captured by PC1–PC5 is shown in the supporting information (see figure S2). Although 95.8% of the variance in the dataset is captured by the first two PCs, there is no observable boundary between solutions with different conductivities in the score plot. In particular, in solutions with high conductivities of 1345, 1733, and 2325 μS cm^{-1}, most of the data points are overlapped. The results indicate that the features selected by PCA cannot discriminate the conductivity of solutions although 95.8% of the variance within the dataset is retained. We also check the score plots of higher PCs. The score plot constructed by PC3 and PC4 is shown in the supporting information (see figure S3). It shows that more data points with different conductivities are overlapped, as compared with the score plot of PC1 and PC2. We further analyze the PCA results by using the loading plots of PC1 and PC2 to understand the construction of the PCs. Figure 6(b) shows the typical spectrum and loadings at each wavelength. The results show that the emission lines of H_α, H_β, O, and OH dominate the loading plots of both PC1 and PC2. The characteristics of loadings indicate that PCA performs transformation mainly depending on the characteristics emission lines in the spectra. We conclude that PCA primarily identifies the

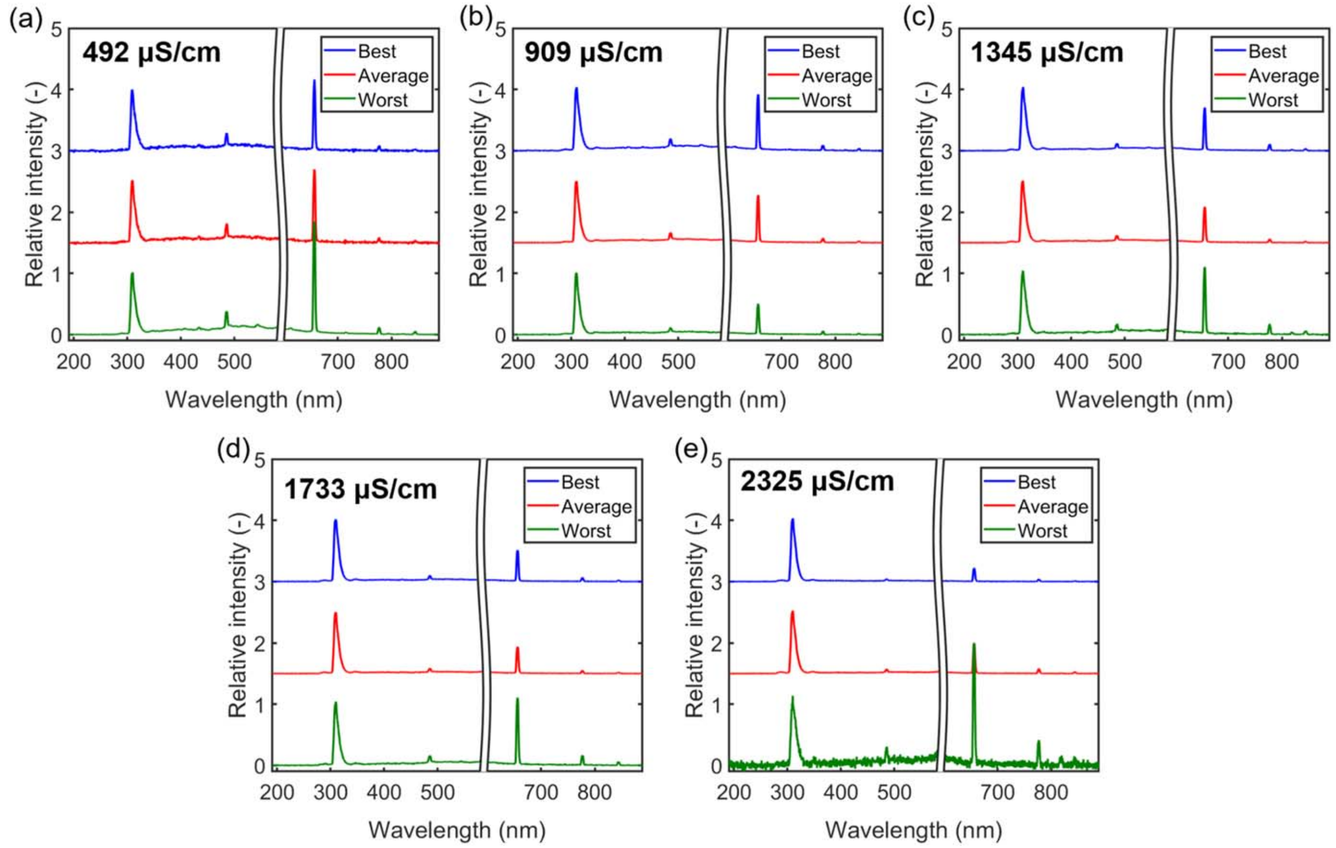

Figure 5. Spectra of the best, average, and the worst accuracy from the fitted line using a single emission line in solutions with conductivities of (a) 492, (b) 909, (c) 1345, (d) 1733, and (e) 2325 μS cm^{-1}.

variance among datasets with each conductivity rather than the patterns for different conductivities.

The above results show the limitation of PCA for its algorithm. It interprets the data depending on the variance in the dataset without considering that within the dataset, that is, conductivity in this study. In plasma systems that show highly non-uniform characteristics, the PCA algorithm selects the variance caused by the non-uniform nature of plasma behaviors rather than by the condition or information of interest. Further, the PCs identified by PCA are constructed by linear transformation of the spectra. When handling datasets with nonlinear patterns, the results may be distorted. Because of these limitations of PCA, we further test a machine learning algorithm allowing for supervised learning with nonlinear algorithms to analyze the spectra.

3.3. Quantification of solution conductivity using an ANN

In this study, a supervised machine learning algorithm ANN is tested because ANN imposes fewer restrictions on input data and affords higher flexibility for performing linear and nonlinear calculations. We use ANN for quantifying the conductivity of solutions. Figures 7(a)–(d) show the parity plots of the measured values from the solution conductivity probe and the predicted values using a single emission line, ANN with zero hidden layers, shallow ANN, and deep ANN, respectively. Table 2 lists the corresponding MSEs and RMSEs for each case. For predictions with a single emission line, the H_α emission line integrated with first-order linear regression is used, as discussed in the previous section. We refer to this framework as the conventional method. For this purpose, 40k spectra are mapped in the parity plot. For predictions using ANNs, spectra with 1909 pixels are provided to the input layer. We use 95% of the spectra in the dataset and separate them into five groups. Of these, four (with 30.4k spectra) are used for training and one (with 7.6k spectrums) is used for validation. Several frameworks are constructed. In the ANN with zero hidden layers, the signals from the input layer are directly delivered to the output layer without a hidden layer between. The predictions are calculated through a linear combination of the intensities of each pixel. The goal is to highlight the improvement afforded by multivariate analysis with the simplest network structure compared with the conventional method. The shallow ANN contains one hidden layer with 50 neurons. The ReLU nonlinear activation function is applied in each neuron. With the introduction of the hidden layer, the framework can perform nonlinear transformation. The deep ANN contains three hidden layers with 2000, 50, and 10 neurons in the 1st, 2nd, and 3rd hidden layers, respectively. ReLU is applied to each neuron in the hidden layers. The deep ANN can handle the datasets with higher complexity. We also use the dropout and early stopping techniques to prevent the training process from overfitting in the deep ANN. Specifically, a 10% dropout rate in

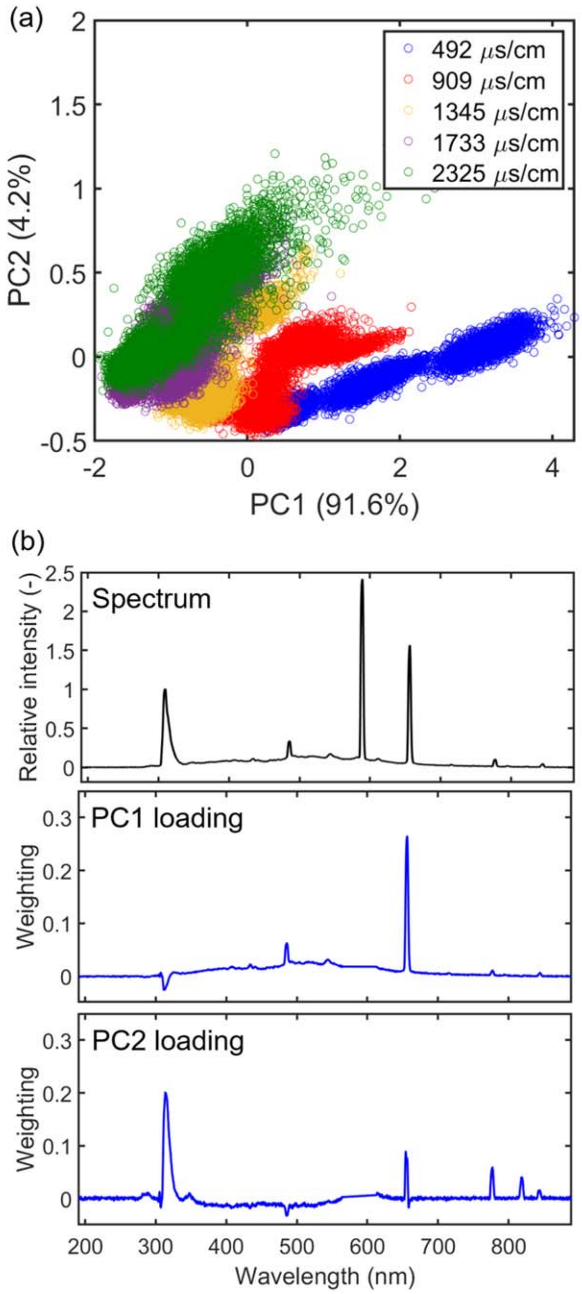

Figure 6. (a) PCA scores plot of PC1 and PC2. The points show the scores of each data in the dataset. (b) Typical OES and loading plots of PC1 and PC2.

the first hidden layer and tolerance of 80 epochs for early stopping are applied.

The results show that the application of an ANN significantly improves the accuracy of conductivity predictions. Compared to the conventional method using a single emission line, the MSE decreases to ∼25% when using ANN with zero hidden layers. With the further introduction of one hidden layer (shallow ANN), the accuracy greatly improves in the parity plot (figure 7(c)) with the MSE decreasing by more than two orders of magnitude with respect to the conventional method (using a single emission line for the analysis). In this case, we observe that the accuracy of the validation data is lower than that of the training data in the parity plot. The MSE for the validation data is nearly twice higher than that of the training data. These results suggest that the framework encounters overfitting. We note that the dropout and early stopping techniques are not implemented during the training of the shallow ANN. Further tests with the deep ANN reveal that it has the best performance. First, we observe that all data points cluster to the reference line in the parity plot (figure 7(d)). The MSEs in both the training and validation data improve significantly by three orders of magnitude to 109 and 146, respectively, as compared with the method using a single emission line. We note that the mean absolute percentage error of the prediction is lower than 1%, indicating that the prediction accuracy is higher than 99%. The coefficient of determination, R^2 values, for each case are tabulated in the supporting information (see table S1). It shows that for the deep ANN, the R^2 values for the training and validation data are 0.999 98 and 0.999 89, respectively. With the uses of dropout and early stopping, the MSE of the training and validation data shows similar values, implying that the overfitting issue is mitigated.

The improvement in accuracy achieved using the ANN shows the advantage of multivariate analysis for the OES of plasma. For the ANN with zero hidden layers, a simple linear combination of intensities at all wavelengths in the spectrum is performed. The accuracy is higher than that obtained with a single emission line, indicating that solution conductivity can be better predicted using multiple emission lines. A rather large MSE suggests that the framework is not good enough to provide a reliable conductivity prediction. With the introduction of the hidden layer, much lower MSEs in the training and validation data are obtained. This advancement is achieved not only because of the introduction of more parameters to fit in the framework but also because of the use of nonlinear transformations provided in each neuron. It is straightforward that providing more parameters leads to higher accuracy. However, doing so also increases the risk of overfitting, as shown in the shallow ANN. The nonlinear transformation is performed by introducing an activation function that considers the nonlinear relationship. We note that the introduction of nonlinear activation function brings the most significant improvement on accuracy. The improvement achieved using the hidden layer indicates the nonlinear behaviors between the emission intensities at different wavelengths and the solution conductivity. Stacking more hidden layers, as in the deep ANN, provides the best results. The deep ANN can handle datasets with high complexity. Therefore, it is a better framework for handling datasets with features that are hard to analyze. We also note that using the deep ANN significantly increases the number of parameters to be optimized. Such an increase makes the use of this experimental platform allowing for rapid acquisition of emission spectra highly desirable.

Overfitting is a common problem when using an ANN. It is defined as a situation in which the MSE obtained on the training set is much lower than that obtained on the validation set. Therefore, it can be identified rather easily. In the deep ANN, we implement the dropout and early stopping techniques to mitigate overfitting. A 10% dropout rate is applied in the first hidden layer and early stopping with a tolerance of 80

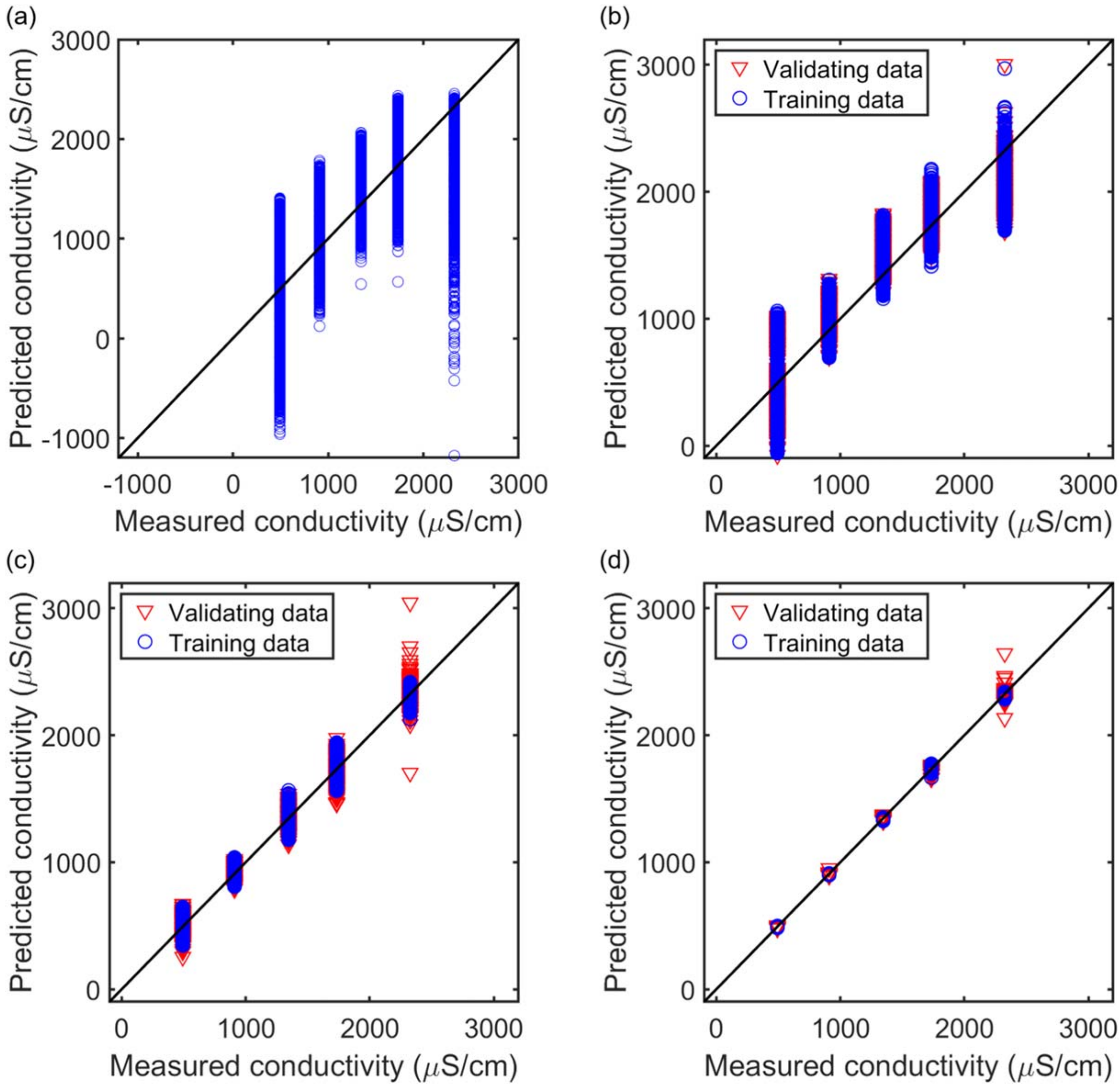

Figure 7. Parity plots of conductivity measured using a solution conductivity probe and conductivity predicted using (a) a single emission line, (b) ANN with zero hidden layers, (c) shallow ANN with 50 neurons, and (d) deep ANN with three hidden layers containing 2000, 50, and 10 neurons in the 1st, 2nd, and 3rd hidden layers, respectively. In the ANNs, the blues points and red points indicate the training and validation data, respectively.

Table 2. MSE and RMSE of the training and validation data using single emission line, ANN with zero hidden layers, shallow ANN, and deep ANN. The units of the MSE and RMSE are $\mu S^2/cm^2$ and $\mu S/cm$, respectively.

	Training MSE (RMSE)	Validation MSE (RMSE)
Single emission line	2.31×10^5 (1.52×10^3)	
ANN with zero hidden layers	5.54×10^4 (235)	5.54×10^4 (235)
Shallow ANN	1.18×10^3 (34.4)	2.15×10^3 (46.4)
Deep ANN	109 (10.4)	146 (12.1)

epochs is demonstrated. In the deep ANN, the validation MSE is around four times higher than the training MSE without implementing dropout and early stopping techniques (not shown). By contrast, when implementing these techniques, the difference between the training and validation MSE is ∼15%, suggesting that more robust parameters are trained.

3.4. Test results using a deep neural network

We finally examine the test results obtained using the trained deep ANN discussed in the previous section. Figure 8 shows the parity plot results for the test data. 5% of the spectra (2k spectra) in the dataset of each solution conductivity are used as the test set. We note that the test set is never seen

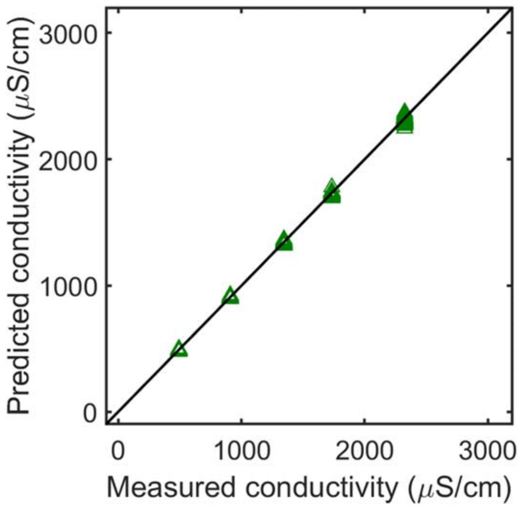

Figure 8. Parity plot of conductivity measured using a solution conductivity probe and conductivity predicted using the trained deep ANN. The green points indicate the test data.

during the training processes. It provides an unbiased and independent evaluation of the training results. We apply the test set to the trained deep ANN. The results show that the data points are highly correlated with the reference line in the parity plot with an MSE of 138 ($\mu S^2/cm^2$) and RMSE of 11.7 ($\mu S/cm$). The value of the test MSE is similar to that of the training and validation MSE. It implies that the overfitting issue is resolved during training and does not occur. Further, the results indicate that the deep ANN is well trained and that this framework can be applied to quantify conductivity by OES using plasma ignited in the experimental and solution conditions. The time required for solution conductivity prediction with each spectrum using the trained deep ANN is within 1 μs, which is sufficiently fast for typical for real-time monitoring. We note that to extend the current ANN results to other experimental or solution conditions, obtaining a proper dataset for the new experimental set and training is definitely required. The selection of algorithm and its detailed scheme, such as a shallow versus deep ANN, dropout and early stopping, is expected to be applicable to OES analysis for other systems. In addition, several machine learning approaches, such as transfer learning and active learning, can potentially be utilized to assist or facilitate the training process for new systems or different conditions.

4. Conclusion

This study discusses machine learning of OES of plasma in aqueous solutions. We develop a plasma OES generation platform that enables the efficient collection of the OES and use several machine learning algorithms to analyze the plasma. We quantify the conductivity of solutions with various pH values using the spectra. In total, 40k spectra are collected from the platform and applied to machine learning.

First, we examine the capability of quantifying conductivity using the relative intensity of H_α/OH, which is referred to as the conventional method. Owing to the behaviors of plasma and that its optical emission is complicated and non-uniform spatially and temporally, the conductivity quantification ability is poor. Then, we use PCA and ANN to perform multivariate analysis of the spectra. The PCA results show that the data points are overlapped in the score plot constructed using PC1 and PC2 although 95.8% of the variance is retained. PCA identifies the variance among datasets with each conductivity rather than the patterns for different conductivities. In addition, the relationship between the plasma and solution conductivity may be nonlinear. Such results indicate that PCA cannot discriminate the conductivity and has limitations. Several ANN structures, namely, ANN with zero hidden layers, shallow ANN, and deep ANN, are constructed. The use of ANNs improves accuracy by four times in terms of MSE compared with the conventional method, thereby showing the advantages of multivariate analysis. Further, deep ANN improves the accuracy in terms of MSE by three orders of magnitude. This improvement implies that deep ANN considers the nonlinear behavior of plasma. The dropout and early stopping techniques are implemented to mitigate overfitting. The test set, which is independent of the training processes, is applied to the deep ANN. The value of the test MSE is similar to that of the training and validation MSEs, indicating the framework is robust. The application of a deep ANN with a large number of parameters makes the use of the platform for efficient acquisition of spectra highly desirable. Utilizing machine learning to analyze OES offers an efficient and accurate route to quantify solution conductivity in plasmas ignited in aqueous solution. We believe that this methodology, analyzing OES with machine learning algorithms, can be further applied to analyze not only solution conductivity but also various properties such as certain species identification or quantification for different plasma systems in various applications. This study using machine learning to analyze OES offers a new route for characterizing plasma.

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